

TOURNAMENT OF TOWNS

31st Tournament • Academic Year 2009–2010

Fall (Autumn) Round • 18 October 2009 • Junior (Grades 8–9) • Basic (O-Level)

Score is based on the three problems with the best results.

Problem 1. [3 points]

Can a square be cut into 9 squares and painted so that there are 1 white, 3 gray, and 5 black squares, with squares of the same color being equal and squares of different colors being unequal?

(N.I. Avilov)

Problem 2. [4 points]

There are 40 weights with masses of 1 g, 2 g, ..., 40 g. 10 weights of even mass are chosen and placed on the left pan of the scales. Then, 10 weights of odd mass are chosen and placed on the right pan of the scales. The scales are in equilibrium. Prove that on one of the pans there are two weights with a mass difference of 20 g.

(V.V. Proizvolov)

Problem 3. [4 points]

A cardboard circle with a radius of 5 cm is on the table. Petya applies cardboard squares with a side length of 5 cm to the circle from the outside as long as possible, so that the following conditions are met: 1) one vertex of each square lies on the circle's boundary; 2) the squares do not overlap; 3) each subsequent square touches the previous one vertex to vertex. Determine how many squares Petya can lay out and prove that the last and first squares will also touch vertex to vertex.

(A.V. Shapovalov)

Problem 4. [5 points]

A seven-digit code consisting of seven different digits is called good. The safe's password is a good code. It is known that the safe will open if a good code is entered and at some place the code's digit matches the corresponding digit of the password. Can the safe be guaranteed to be opened faster than in 7 attempts?

(D.V. Baranov)

Problem 5. [5 points]

2000 people registered on the new site. Each invited 1000 people to be their friends. Two people are considered friends if and only if each of them invited the other to be their friend. What is the smallest number of pairs of friends that could have formed?

(A.Yu. Evnin)